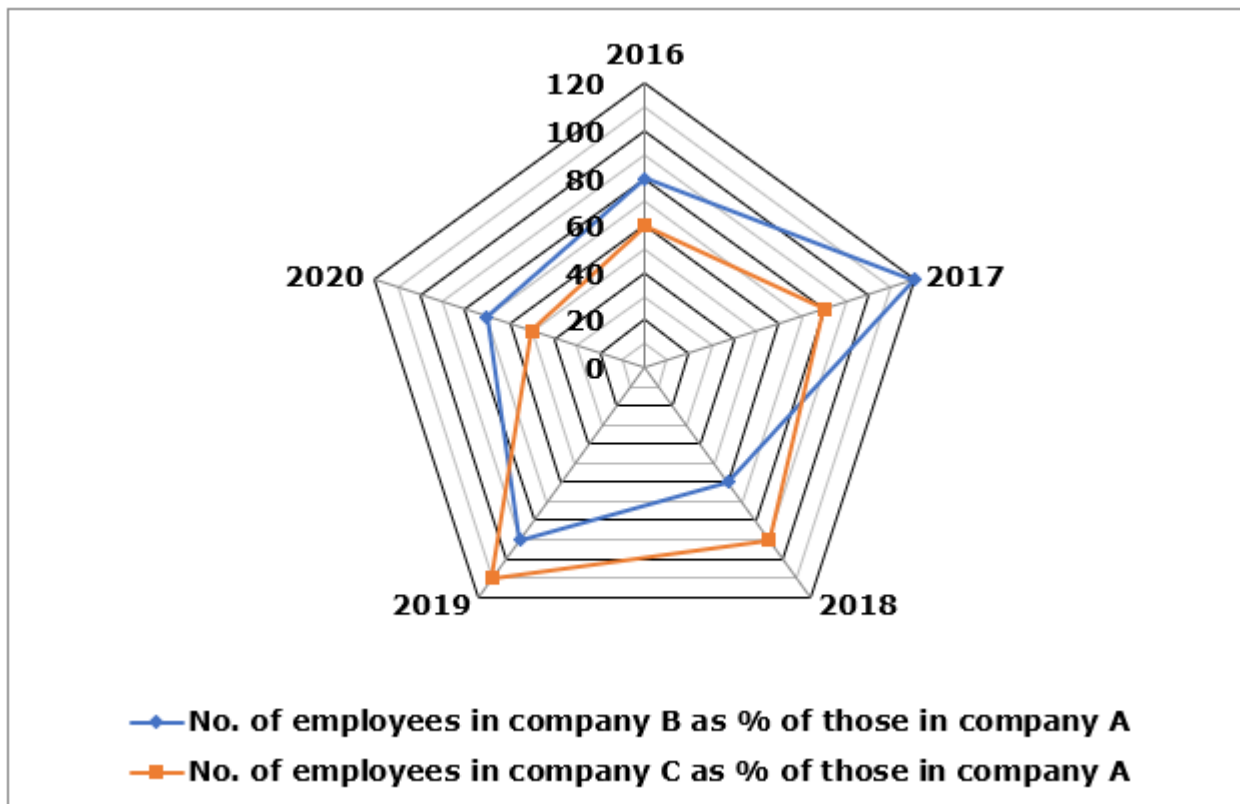


1. Questions

Study the following data carefully and answer the questions:

The given radar graph shows below is related to the number of employees in three companies A, B, and C in five different years i.e. 2016, 2017, 2018, 2019, and 2020.



If the difference between the number of employees in companies B and C in 2017 is 20 and the difference between the number of employees in company A in 2017 and 2018 is 10, then which of the following cannot be a possible number of employees in company C in 2018?

- 45
- 72
- 63
- Both (b) and (c)
- All (a), (b) and (c)

@QuizByMickey

2. Questions

If the average number of employees in companies A, B, and C in 2020 is 88, the number of employees in company A is decreased by 25% from 2020 to 2021, and the number of employees in company B is increased by 25% from 2020 to 2021 and the number of employees in company C is increased by 20% from 2020 to 2021, then find the average number of employees in companies A, B, and C in 2021?

- 81
- 93

- c. 89
- d. 91
- e. 87

3. Questions

If the difference between the number of employees in companies A and B in 2018 is 24 and the number of employees in company B is increased by 100% from 2018 to 2019, then find the number of employees in company C increased by how much percent (approximately) from 2018 to 2019?

- a. 81%
- b. 63%
- c. 90%
- d. 72%
- e. 54%

4. Questions

If the number of employees in Company A in 2016 is P, which is 12.5% more than those in Company A in 2019, then the difference between the number of employees in Company C in 2016 and those in Company C in 2019 will be Q.

Which of the following pairs of (P, Q) satisfies the given condition?

I: (90, 45)

II: (135, 51)

III: (45, 27)

- a. Only II
- b. Only I and II
- c. Only III
- d. Only I
- e. Only II and III

@QuizByMickey

5. Questions

The ratio of the number of employees in Company A in 2019 to those in Company A in 2020 is 2:3. In 2019, the ratio of the number of employees in Company C to that in company D is 11:6. In 2020, the ratio of the number of employees in company C to that in company D is 5:6. If the number of employees in company D is increased/decreased by M% from 2019 to 2020, then which of the following statement(s) about M is/are true?

I: $(M + 21)$ is a prime number.

II: $(M + 75)$ is a perfect square.

III: $(M - 15)$ is divisible by 9.

- a. Only I and III
- b. All are true
- c. Only I
- d. Only II
- e. Only II and III

6. Questions

If the number of employees in company A in 2016, 2017, 2018, 2019, and 2020 are 70, 60, 80, 50, and 90 respectively, then in which two years will the number of employees in company B be equal?

- a. 2017 and 2020
- b. 2016 and 2019
- c. 2018 and 2019
- d. None of these
- e. 2017 and 2018

7. Questions

Study the following data carefully and answer the questions:

The given table chart shows below is related to the amount invested by five persons i.e. A, B, C, D, and E in a bank in three years 2021, 2022, and 2023.

Persons	Amount invested in 2021 (in ₹)	The ratio of the amount invested in 2021 to that in 2022	The ratio of the amount invested in 2021 to that in 2023
A	$(X - 1000)$	4:5	8:5
B	$(X + Y)$	3:1	5:4
C	$(Y + 500)$	5:8	5:3
D	$(Z + 500)$	3:2	3:4
E	$(X + Z)$	5:2	10:11

Notes:

- I).** The amount invested by A in 2022 is ₹2500 more than that in 2023.
- II).** The difference between the amount invested by B in 2021 and 2022 is ₹5000.
- III).** The ratio of the amount invested by D in 2021 to that by E in 2021 is 1:2.

If the rate of interest in 2021 and 2022 is 4% and 6% respectively, then find the difference between the total simple interest received by B in 2021 and 2022 together and the total simple interest received by C in 2021 and 2022 together? (The amount invested in 2021 is not withdrawn)

- a. ₹302
- b. Can't be determined
- c. ₹322
- d. ₹312
- e. ₹332

8. Questions

If the total interest received by given all five persons together on the amount invested in 2021 only is ₹3500 and the total interest received by given all five persons together on the amount invested in 2022 only is ₹945, then find the ratio of the rate of interest in 2021 to that in 2022?

- a. 4:3
- b. 5:2
- c. 2:1
- d. 3:1
- e. 5:3

9. Questions

If the rate of interest in 2021, 2022, and 2023 were 7%, 5%, and 4% respectively, then the ratio of the simple interest received by E in 2021, 2022, and 2023 would have been P:Q:R. (The amounts invested in 2021 and 2022 are not withdrawn)

Find the roots of the equation $[Pa^2 - (2Q + 5)a + R = 0]$?

- a. $2/7, 5/7$
- b. $2, 5/7$
- c. $2/7, 5$
- d. $2, 5$
- e. None of these

@QuizByMickey

10. Questions

The amount invested by F in the bank in 2021, 2022 and 2023 was ₹(Y + Z), ₹3.5(X - Z) and ₹(X - Y) respectively. The rates of interest in 2021, 2022, and 2023 were $(Z/1000)\%$, $[(X/1000) + 1]\%$ and $(2Z/1000)\%$ respectively. What would be the total simple interest received by F from all three investments together? (The amount invested in 2021 is withdrawn after 2021 and the amount invested in 2022 is withdrawn after 2022)

- a. ₹620
- b. ₹540
- c. ₹670
- d. ₹640
- e. ₹500

11. Questions

If the rate of interest in 2023 is $M\%$ and the simple interest received by A on the amount invested in 2023 only is ₹44.8 more than that received by C, then find the total simple interest received by B and D together on the amount invested in 2023 only?

- a. ₹(X – Z – 232)
- b. ₹(X – Z – 768)
- c. ₹(X – Z – 544)
- d. ₹(X – Z – 456)
- e. ₹(X – Z – 1000)

12. Questions

If the amount invested by A in 2024 is ₹___ and the amount invested by D in 2024 is ₹___, then the ratio of the total amount invested by A over the four years from 2021 to 2024 to that by D will be 16: 15.

Which of the following cannot be filled in the blanks in the same order?

I: 4500, 1500

II: 3000, 1000

III: 6000, 2000

- a. Only II and III
- b. Only I and III
- c. Only II
- d. Only III
- e. Only I and II

@QuizByMickey

13. Questions

Study the following data carefully and answer the questions:

Two inlet pipes i.e. A and B and two outlet pipes i.e. C and D are connected to the tank.

The ratio of the efficiency of pipe A to that of pipe B is 2:3 and the ratio of the efficiency of pipe C to that

of pipe D is 5:4. Pipes A and C are opened together. After 64 minutes, when $66\frac{2}{3}\%$ part of the tank is filled, pipes A and C are closed and pipes B and D are opened together. Pipes B and D together filled the remaining part of the tank in $8\frac{8}{9}$ minutes.

If the efficiency of pipe B were 60% more than its original efficiency and the ratio of the new efficiency of pipe B to that of pipe C were 5:2, then the new efficiency of pipe C would have been what per cent more than its original efficiency?

- a. 21%
- b. 35%
- c. 32%
- d. 16%
- e. 28%

14. Questions

If pipes A and D together can fill the tank in 'p' minutes and pipes B and C together can fill the tank in (p – q) minutes, then which of the following statement(s) is/are true?

I: 'p' and 'q' are co-prime numbers with each other.

II: The difference of 'p' and 'q' is a multiple of 8.

III: The value of (2p + 5q) is a multiple of 13.

- a. Only II and III
- b. Only II
- c. Only I and III
- d. Only III
- e. Only I

15. Questions

Pipes A, B, C and D are opened together. After how many minutes, pipes A and D should be closed, so that the remaining part of tank is filled by pipes B and C together in 4.4 minutes?

- a. 15 minutes
- b. 21 minutes
- c. 12 minutes
- d. 18 minutes
- e. 24 minutes

16. Questions

Study the following data carefully and answer the questions:

Two series I and II are given below and the both the series follow the same pattern:

Series I: 11, 12, 22, 69, 272, 1365

Series II: 6, X, (X + Y), (X + Y + Z)

Which of the following statement(s) is/are true?

P: The value of $(Z - 3X)^Y$ is 7766.

Q: The value of $(Z - X - Y)$ is a prime number.

R: $2(X + Y) < Z$

- a. Only R
- b. Only P and Q
- c. Only Q and R
- d. None is true
- e. Only Q

17. Questions

If a series is formed, whose 1st term is $(Z - 2X - 2Y)$ and it follows the same pattern as in the given series, then find the 5th term of the series?

- a. 64
- b. 108
- c. 80
- d. 72
- e. 96

18. Questions

Study the following data carefully and answer the questions:

Three series I, II and III are given below with missing terms P, Q and R in series I, II and III respectively.

I: 5, 13, 29, 61, P, 253

II: 1200, 839, Q, 381, 260, 211

III: 10, 30, 75, 150, 225, R

Find the ratio between P, Q and R?

- a. 5: 11: 9
- b. 15: 22: 18
- c. 1: 3: 2

- d. 5: 22: 9
- e. None of these

19. Questions

Find the value of $(Q - P^{1/3} - R^{1/2})$ is divisible by which of the following?

- a. 20
- b. 103
- c. 15
- d. Both (a) and (b)
- e. None of these

20. Questions

Two series I and II are given below with the missing term P in series I, the wrong term Q in series II and the term R, which will replace the wrong term of series II.

I: 21, ?, 57, 90, 141, 216

II: 21, 44, 73, 104, 140, 182

Find the average of P, Q and R?

- a. 115.67
- b. 85.67
- c. 105.67
- d. 96.67
- e. 123.67

Explanations:**1. Questions**

@QuizByMickey

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

Answer: E

Let the number of employees in company A in 2017 be 5a

So, the number of employees in company B in 2017 = $(120/100) * 5a = 6a$

The number of employees in company C in 2017 = $(80/100) * 5a = 4a$

According to the question,

$$6a - 4a = 20$$

$$a = 10$$

The number of employees in company A in 2017 = $5 * 10 = 50$

The difference between the number of employees in company A in 2017 and 2018 is 10.

So, the number of employees in company A in 2018 will either be $50 + 10 = 60$ or $50 - 10 = 40$

If the number of employees in company A in 2018 = 60

Then, the number of employees in company C in 2018 = $(90/100) * 60 = 54$

If the number of employees in company A in 2018 = 40

Then the number of employees in company C in 2018 = $(90/100) * 40 = 36$

So, the number of employees in company C in 2018 will either be 54 or 36.

Therefore, the calculated values don't match the values given in the options,

So, option E is correct.

@QuizByMickey

2. Questions

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

Answer: C

Let the number of employees in company A be 100a

So, the number of employees in company B = $100a * (70/100) = 70a$

The number of employees in company C = $100a * (50/100) = 50a$

The ratio of the number of employees in companies A, B, and C respectively in 2020 = 100a: 70a: 50a = 10:7:5

The total number of employees in companies A, B, and C together in 2020 = $3 * 88 = 264$

The number of employees in company A in 2020 = $264 * [10/(10 + 7 + 5)] = 120$

The number of employees in company B in 2020 = $264 * [7/(10 + 7 + 5)] = 84$

The number of employees in company C in 2020 = $264 * [5/(10 + 7 + 5)] = 60$

The number of employees in company A in 2021 = $(75/100) * 120 = 90$

The number of employees in company B in 2021 = $(125/100) * 84 = 105$

The number of employees in company C in 2021 = $(120/100) * 60 = 72$

The required average = $(90 + 105 + 72)/3 = 267/3 = 89$

3. Questions

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

Answer: B

@QuizByMickey

Let the number of employees in company A in 2018 be $100x$

So, the number of employees in company B in 2018 = $100x * (60/100) = 60x$

Then, the number of employees in company C in 2018 = $100x * (90/100) = 90x$

The ratio between the number of employees in companies A, B, and C respectively in 2018 = $100x : 60x : 90x = 10:6:9$

According to the question,

The number of employees in company A in 2018 = $24 * [10/(10 - 6)] = 60$

The number of employees in company B in 2018 = $60 * (6/10) = 36$

The number of employees in company C in 2018 = $60 * (9/10) = 54$

The number of employees in company B in 2019 = $(200/100) * 36 = 72$

The number of employees in company A in 2019 = $72 * (100/90) = 80$

The number of employees in company C in 2019 = $(110/100) * 80 = 88$

The required percentage = $[(88 - 54)/54] * 100 = 62.96\% \sim 63\%$ (approx.)

4. Questions

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

Answer: A

From I:

The number of employees in company A in 2016 = $P = 90$

The number of employees in company A in 2019 = $90 * (100/112.5) = 80$

The number of employees in company C in 2016 = $(60/100) * 90 = 54$

The number of employees in company C in 2019 = $(110/100) * 80 = 88$

So, $Q = 88 - 54$

$Q = 34$

Therefore, the pair of (P, Q) given in I doesn't satisfy the given condition.

From II:

@QuizByMickey

The number of employees in company A in 2016 = P = 135

The number of employees in company A in 2019 = $135 * (100/112.5) = 120$

The number of employees in company C in 2016 = $(60/100) * 135 = 81$

The number of employees in company C in 2019 = $(110/100) * 120 = 132$

So, $Q = 132 - 81$

$Q = 51$

Therefore, the pair of (P, Q) given in II satisfies the given condition.

From III:

The number of employees in company A in 2016 = P = 45

The number of employees in company A in 2019 = $45 * (100/112.5) = 40$

The number of employees in company C in 2016 = $(60/100) * 45 = 27$

The number of employees in company C in 2019 = $(110/100) * 40 = 44$

So, $Q = 44 - 27$

$Q = 17$

Therefore, the pair of (P, Q) given in III doesn't satisfy the given condition.

Hence, the pair of (P, Q) given in II only satisfies the given condition.

5. Questions

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

@QuizByMickey

Answer: C

Let the number of employees in company A in 2019 and 2020 be $20x$ and $30x$ respectively.

The number of employees in company C in 2019 = $(110/100) * 20x = 22x$

The number of employees in company C in 2020 = $(50/100) * 30x = 15x$

According to the question,

The number of employees in company D in 2019 = $22x * (6/11) = 12x$

So, the number of employees in company D in 2020 = $15x * (6/5) = 18x$

And, $M = [(18x - 12x)/12x] * 100$

$M = 50\%$

Statement I:

The value of $(M + 21) = 50 + 21 = 71$

Therefore, 71 is a prime number.

So, statement I is true.

Statement II:

The value of $(M + 75) = 50 + 75 = 125$

Therefore, 125 is a perfect cube but not a perfect square.

So, statement II is not true.

Statement III:

The value of $(M - 15) = 50 - 15 = 35$

Therefore, 35 is not divisible by 9.

So, statement III is not true.

Thus, only I satisfy the condition.

Hence, option C

6. Questions

Year	No. of employees in company B as % of those in Company A	No. of employees in company C as % of those in Company A
2016	80%	60%
2017	120%	80%
2018	60%	90%
2019	90%	110%
2020	70%	50%

@QuizByMickey

Answer: D

We know that, the numbers of employees in company A in 2016, 2017, 2018, 2019, and 2020 are 70, 60, 80, 50, and 90 respectively.

So, the number of employees in company B in 2016 = $(80/100) * 70 = 56$

The number of employees in company B in 2017 = $(120/100) * 60 = 72$

The number of employees in company B in 2018 = $(60/100) * 80 = 48$

The number of employees in company B in 2019 = $(90/100) * 50 = 45$

And the number of employees in company B in 2020 = $(70/100) * 90 = 63$

Therefore, the numbers of employees in company B in each of the given years are different.

Hence, option D is correct.

7. Questions

According to the table,

The amount invested by A in 2022 = $(X - 1000) * (5/4) = ₹[(5X - 5000)/4]$

The amount invested by A in 2023 = $(X - 1000) * (5/8) = ₹[(5X - 5000)/8]$

According to the Note I,

$$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$$

$$(10X - 10000 - 5X + 5000) = 20000$$

$$5X - 5000 = 20000$$

$$X = 25000/5$$

$$X = 5000$$

According to the Note II,

The amount invested by B in 2021 = $5000 * 3/(3 - 1) = ₹7500$

$$(X + Y) = 7500$$

$$Y = 7500 - X$$

$$Y = 7500 - 5000$$

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

@QuizByMickey

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: D

We know that, the amount invested by B in 2021 = ₹7500

And the amount invested by B in 2022 = ₹2500

The SI received by B in 2021 and 2022 together = $[(7500 * 4)/100] + [(7500 + 2500) * 6]/100 = 300 + \{100 * 6\} = 300 + 600 = ₹900$

We know that, the amount invested by C in 2021 = ₹3000

And the amount invested by C in 2022 = ₹4800

The SI received by C in 2021 and 2022 together = $[(3000 * 4)/100] + [(3000 + 4800) * 6]/100 = 120 + \{78 * 6\} = 120 + 468 = ₹588$

The required difference = $900 - 588 = ₹312$

8. Questions

According to the table,

The amount invested by A in 2022 = $(X - 1000) * (5/4) = ₹[(5X - 5000)/4]$

The amount invested by A in 2023 = $(X - 1000) * (5/8) = ₹[(5X - 5000)/8]$

According to the Note I,

$$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$$

$$(10X - 10000 - 5X + 5000) = 20000$$

$$5X - 5000 = 20000$$

$$X = 25000/5$$

$$X = 5000$$

According to the Note II,

The amount invested by B in 2021 = $5000 * 3/(3 - 1) = ₹7500$

$$(X + Y) = 7500$$

$$Y = 7500 - X$$

$$Y = 7500 - 5000$$

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

@QuizByMickey

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: B

Let the rate of interest in 2021 be 'R'%.

The total amount invested by all five persons together in 2021 only = $(4000 + 7500 + 3000 + 4500 + 9000)$
 = ₹28000

Simple interest = $PNR/100$

$$(28000 * R * 1)/100 = 3500$$

$$R = 12.5$$

The rate of interest in 2021 = 12.5%

Let the rate of interest in 2022 be 'r'%.

The total amount invested by all five persons together in 2022 only = $(5000 + 2500 + 4800 + 3000 + 3600)$
 = ₹18900

$$(18900 * r * 1)/100 = 945$$

$$r = 5$$

The rate of interest in 2022 = 5%

The required ratio = $R:r = 12.5:5 = 5:2$

9. Questions

According to the table,

The amount invested by A in 2022 = $(X - 1000) * (5/4) = ₹[(5x - 5000)/4]$

The amount invested by A in 2023 = $(X - 1000) * (5/8) = ₹[(5X - 5000)/8]$

According to the Note I,

$$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$$

$$(10X - 10000 - 5X + 5000) = 20000$$

$$5X - 5000 = 20000$$

$$X = 25000/5$$

$$X = 5000$$

According to the Note II,

$$\text{The amount invested by B in 2021} = 5000 * 3/(3 - 1) = ₹7500$$

$$(X + Y) = 7500$$

$$Y = 7500 - X$$

$$Y = 7500 - 5000$$

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: B

We know that, the amount invested by E in 2021 = ₹9000

The amount invested by E in 2022 = ₹3600

And the amount invested by E in 2023 = ₹9900

Simple interest = $PNR/100$

The SI received by E in 2021 = $(9000 * 7 * 1)/100 = ₹630$

The SI received by E in 2022 = $[(9000 + 3600) * 5 * 1]/100 = 126 * 5 = ₹630$

The SI received by E in 2023 = $[(9000 + 3600 + 9900) * 4 * 1]/100 = 225 * 4 = ₹900$

And, P: Q: R = 630: 630: 900

P: Q: R = 7:7:10

Now, from the equation:

$$Pa^2 - (2Q + 5)a + R = 0$$

$$7a^2 - (2 * 7 + 5)a + 10 = 0$$

$$7a^2 - 19a + 10 = 0$$

$$7a^2 - 14a - 5a + 10 = 0$$

$$7a(a - 2) - 5(a - 2) = 0$$

$$a = 2, 5/7$$

10. Questions

According to the table,

$$\text{The amount invested by A in 2022} = (X - 1000) * (5/4) = ₹[(5X - 5000)/4]$$

$$\text{The amount invested by A in 2023} = (X - 1000) * (5/8) = ₹[(5X - 5000)/8]$$

According to the Note I,

$$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$$

$$(10X - 10000 - 5X + 5000) = 20000$$

$$5X - 5000 = 20000$$

$$X = 25000/5$$

$$X = 5000$$

According to the Note II,

$$\text{The amount invested by B in 2021} = 5000 * 3/(3 - 1) = ₹7500$$

$$(X + Y) = 7500$$

$$Y = 7500 - X$$

$$Y = 7500 - 5000$$

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

@QuizByMickey

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: C

The amount invested by F in 2021 = $(Y + Z) = (2500 + 4000) = ₹6500$

The amount invested by F in 2022 = $3.5(X - Z) = 3.5 * (5000 - 4000) = ₹3500$

The amount invested by F in 2023 = $(X - Y) = (5000 - 2500) = ₹2500$

The rate of interest in 2021 = $(Z/1000) = (4000/1000) = 4\%$

The rate of interest in 2022 = $[(X/1000) + 1] = [(5000/1000) + 1] = 6\%$

The rate of interest in 2023 = $(2Z/1000) = (2 * 4000/1000) = 8\%$

The SI received by F in given all three years together = $[(6500 * 4 * 1)/100] + [(3500 * 6 * 1)/100] + [(2500 * 8 * 1)/100]$

$= 260 + 210 + 200 = ₹670$

11. Questions

According to the table,

The amount invested by A in 2022 = $(X - 1000) * (5/4) = ₹[(5X - 5000)/4]$

The amount invested by A in 2023 = $(X - 1000) * (5/8) = ₹[(5X - 5000)/8]$

According to the Note I,

$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$

$(10X - 10000 - 5X + 5000) = 20000$

$5X - 5000 = 20000$

$X = 25000/5$

$X = 5000$

According to the Note II,

The amount invested by B in 2021 = $5000 * 3/(3 - 1) = ₹7500$

$(X + Y) = 7500$

$Y = 7500 - X$

$Y = 7500 - 5000$

@QuizByMickey

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: A

We know that, the amount invested by A in 2023 = ₹2500

And the amount invested by C in 2023 = ₹1800

According to the question,

$$[(2500 * M * 1)/100] - [(1800 * M * 1)/100] = 44.8$$

$$25M - 18M = 44.8$$

$$M = 44.8/7$$

$$M = 6.4$$

The total amount invested by B and D together in 2023 = 6000 + 6000 = ₹12000

So, the total SI received by B and D together on the amount invested in 2023 only = $(12000 * 6.4 * 1)/100$
= ₹768

We check option wise,

$$X = 5000 \text{ \& } Z = 4000$$

$$\text{Option a) } ₹(X - Z - 232) = 5000 - 4000 - 232 = ₹768$$

12. Questions

According to the table,

$$\text{The amount invested by A in 2022} = (X - 1000) * (5/4) = ₹[(5x - 5000)/4]$$

$$\text{The amount invested by A in 2023} = (X - 1000) * (5/8) = ₹[(5X - 5000)/8]$$

According to the Note I,

@QuizByMickey

$$[(5X - 5000)/4] - [(5X - 5000)/8] = 2500$$

$$(10X - 10000 - 5X + 5000) = 20000$$

$$5X - 5000 = 20000$$

$$X = 25000/5$$

$$X = 5000$$

According to the Note II,

$$\text{The amount invested by B in 2021} = 5000 * 3/(3 - 1) = ₹7500$$

$$(X + Y) = 7500$$

$$Y = 7500 - X$$

$$Y = 7500 - 5000$$

$$Y = 2500$$

According to the Note III,

$$(Z + 500)/(X + Z) = 1/2$$

$$(Z + 500)/(5000 + Z) = 1/2$$

$$2Z + 1000 = 5000 + Z$$

$$Z = 4000$$

Persons	The amount invested in 2021 (in ₹)	The amount invested in 2022 (in ₹)	The amount invested in 2023 (in ₹)
A	$(X - 1000) = (5000 - 1000) = 4000$	$4000 * (5/4) = 5000$	$4000 * (5/8) = 2500$
B	$5000 + 2500 = 7500$	$7500 * (1/3) = 2500$	$7500 * (4/5) = 6000$
C	$2500 + 500 = 3000$	$3000 * (8/5) = 4800$	$3000 * (3/5) = 1800$
D	$4000 + 500 = 4500$	$4500 * (2/3) = 3000$	$4500 * (4/3) = 6000$
E	$5000 + 4000 = 9000$	$9000 * (2/5) = 3600$	$9000 * (11/10) = 9900$

Answer: A

The total amount invested by A in 2021, 2022, and 2023 together = $4000 + 5000 + 2500 = ₹11500$

The total amount invested by D in 2021, 2022, and 2023 together = $4500 + 3000 + 6000 = ₹13500$

From I:

When the amount invested by A and D in 2024 are ₹4500 and ₹1500 respectively.

The total amount invested by A in all four years together = $11500 + 4500 = ₹16000$

The total amount invested by D in all four years together = $13500 + 1500 = ₹15000$

The required ratio = $16000 : 15000 = 16 : 15$

So, the values in I satisfied the condition.

From II:

When the amounts invested by A and D in 2024 are ₹3000 and ₹1000 respectively.

The total amount invested by A over the four years together = $11500 + 3000 = ₹14500$

The total amount invested by D over the four years together = $13500 + 1000 = ₹14500$

The required ratio = $14500 : 14500 = 1 : 1$

So, the values in II can't be satisfied the condition.

From III:

When the amount invested by A and D in 2024 are ₹6000 and ₹2000 respectively.

The total amount invested by A over the four years together = $11500 + 6000 = ₹17500$

The total amount invested by D over the four years together = $13500 + 2000 = ₹15500$

The required ratio = $17500 : 15500 = 35 : 31$

So, the values in III can't be satisfied the condition.

Therefore, the values in II and III can't be satisfied the condition.

Hence, option A

13. Questions

WKT, the ratio of the efficiency of pipe A to that of pipe B = 2:3

So, let the time taken by pipe A alone and pipe B alone can fill the tank be '3x' minutes and '2x' minutes respectively.

WKT, the ratio of the efficiency of pipe C to that of pipe D = 5:4

So, let the time taken by pipe C alone and pipe D alone can empty the tank be '4y' minutes and '5y' minutes respectively.

WKT, the time taken by pipes A and C together filled the $66\frac{2}{3}\%$ part of the tank = 64 minutes

So, the time taken by pipes A and C together can fill the whole tank = $64 * (300/200) = 96$ minutes (Note: $66\frac{2}{3}\% \sim 2/3$)

According to the question,

$$(1/3x) - (1/4y) = 1/96$$

$$1/x = (1/32) + (3/4y) \text{ -----}>(1)$$

WKT, the time taken by pipes B and D together filled the $33\frac{1}{3}\%$ part of tank = $8\frac{8}{9}$ minutes = $(80/9)$ minutes

So, the time taken by pipes B and D together can fill the whole tank = $(80/9) * (300/100) = (80/3)$ minutes (Note: $33\frac{1}{3}\% \sim 1/3$)

According to the question,

$$(1/2x) - (1/5y) = 3/80$$

@QuizByMickey

$$1/x = (3/40) + (2/5y) \text{ -----}>(2)$$

Equating equation (1) and (2), we get

$$(1/32) + (3/4y) = (3/40) + (2/5y)$$

$$(3/4y) - (2/5y) = (3/40) - (1/32)$$

$$(15 - 8)/20y = (12 - 5)/160$$

$$y = 8$$

The value of y, apply on equation (1),

$$1/x = (1/32) + (3/4y)$$

$$1/x = (1/32) + (3/32)$$

$$x = 8$$

The time taken by pipe A alone can fill the tank = $3 * 8 = 24$ minutes

The time taken by pipe B alone can fill the tank = $2 * 8 = 16$ minutes

The time taken by pipe C alone can empty the tank = $4 * 8 = 32$ minutes

The time taken by pipe D alone can empty the tank = $5 * 8 = 40$ minutes

Answer: E

WKT, the time taken by pipe B can fill the tank with its original efficiency = 16 minutes

So, the time taken by pipe B will fill the tank with its new efficiency = $16 * (100/160) = 10$ minutes

Since, the ratio of the new efficiency of pipe B to that of pipe C = 5:2

So, the time, in which pipe C will empty the tank with its new efficiency = $10 * (5/2) = 25$ minutes

WKT, the time taken by pipe C can empty the tank with its original efficiency = 32 minutes

The required percentage = $[(32 - 25)/25] * 100 = 28\%$

14. Questions

WKT, the ratio of the efficiency of pipe A to that of pipe B = 2:3

So, let the time taken by pipe A alone and pipe B alone can fill the tank be '3x' minutes and '2x' minutes respectively.

WKT, the ratio of the efficiency of pipe C to that of pipe D = 5:4

So, let the time taken by pipe C alone and pipe D alone can empty the tank be '4y' minutes and '5y' minutes respectively.

WKT, the time taken by pipes A and C together filled the $66\frac{2}{3}\%$ part of the tank = 64 minutes

So, the time taken by pipes A and C together can fill the whole tank = $64 * (300/200) = 96$ minutes (Note: $66\frac{2}{3}\% \sim 2/3$)

According to the question,

$$(1/3x) - (1/4y) = 1/96$$

$$1/x = (1/32) + (3/4y) \text{ -----}>(1)$$

WKT, the time taken by pipes B and D together filled the $33\frac{1}{3}\%$ part of tank = $8\frac{8}{9}$ minutes = $(80/9)$ minutes

So, the time taken by pipes B and D together can fill the whole tank = $(80/9) * (300/100) = (80/3)$ minutes (Note: $33\frac{1}{3}\% \sim 1/3$)

According to the question,

$$(1/2x) - (1/5y) = 3/80$$

$$1/x = (3/40) + (2/5y) \text{ -----}>(2)$$

Equating equation (1) and (2), we get

$$(1/32) + (3/4y) = (3/40) + (2/5y)$$

$$(3/4y) - (2/5y) = (3/40) - (1/32)$$

$$(15 - 8)/20y = (12 - 5)/160$$

$$y = 8$$

The value of y, apply on equation (1),

$$1/x = (1/32) + (3/4y)$$

$$1/x = (1/32) + (3/32)$$

$$x = 8$$

The time taken by pipe A alone can fill the tank = $3 * 8 = 24$ minutes

The time taken by pipe B alone can fill the tank = $2 * 8 = 16$ minutes

The time taken by pipe C alone can empty the tank = $4 * 8 = 32$ minutes

The time taken by pipe D alone can empty the tank = $5 * 8 = 40$ minutes

Answer: A

WKT, the part of tank filled by pipes A and D together in 1 minute:

$$= (1/24) - (1/40) = (5 - 3)/120 = 1/60$$

@QuizByMickey

So, the time taken by pipe A and D together can fill the tank = $p = 60$ minutes

WKT, the part of tank filled by pipes B and C together in 1 minute = $(1/16) - (1/32) = (2 - 1)/32 = 1/32$

So, the time taken by pipes B and C together can fill the tank = $(p - q) = 32$ minutes

$$\text{And, } q = 60 - 32$$

$$q = 28$$

Statement I:

WKT, both 60 and 28 are multiple of 4.

So, 'p' and 'q' are not co-prime with each other.

Thus, Statement I is not true.

Statement II:

The difference of 'p' and 'q' = $60 - 28 = 32$

So, 32 is a multiple of 8.

Thus, Statement II is true.

Statement III:

The value of $(2p + 5q) = (2 * 60 + 5 * 28) = 120 + 140 = 260$

So, 260 is a multiple of 13.

Thus, Statement III is true.

Therefore, Only II and III are true.

Hence, option A

15. Questions

WKT, the ratio of the efficiency of pipe A to that of pipe B = 2:3

So, let the time taken by pipe A alone and pipe B alone can fill the tank be '3x' minutes and '2x' minutes respectively.

WKT, the ratio of the efficiency of pipe C to that of pipe D = 5:4

So, let the time taken by pipe C alone and pipe D alone can empty the tank be '4y' minutes and '5y' minutes respectively.

WKT, the time taken by pipes A and C together filled the $66\frac{2}{3}\%$ part of the tank = 64 minutes

So, the time taken by pipes A and C together can fill the whole tank = $64 * (300/200) = 96$ minutes (Note: $66\frac{2}{3}\% \sim 2/3$)

According to the question,

$$(1/3x) - (1/4y) = 1/96$$

$$1/x = (1/32) + (3/4y) \text{ -----}>(1)$$

WKT, the time taken by pipes B and D together filled the $33\frac{1}{3}\%$ part of tank = $8\frac{8}{9}$ minutes = $(80/9)$ minutes

So, the time taken by pipes B and D together can fill the whole tank = $(80/9) * (300/100) = (80/3)$ minutes (Note: $33\frac{1}{3}\% \sim 1/3$)

According to the question,

$$(1/2x) - (1/5y) = 3/80$$

$$1/x = (3/40) + (2/5y) \text{ -----}>(2)$$

Equating equation (1) and (2), we get

$$(1/32) + (3/4y) = (3/40) + (2/5y)$$

$$(3/4y) - (2/5y) = (3/40) - (1/32)$$

$$(15 - 8)/20y = (12 - 5)/160$$

$$y = 8$$

The value of y, apply on equation (1),

$$1/x = (1/32) + (3/4y)$$

$$1/x = (1/32) + (3/32)$$

$$x = 8$$

The time taken by pipe A alone can fill the tank = $3 * 8 = 24$ minutes

The time taken by pipe B alone can fill the tank = $2 * 8 = 16$ minutes

The time taken by pipe C alone can empty the tank = $4 * 8 = 32$ minutes

The time taken by pipe D alone can empty the tank = $5 * 8 = 40$ minutes

Answer: D

Let the capacity of the tank = LCM of 24, 16, 32 and 40 = 480 units

So, the part of tank filled by pipe A alone in 1 minute = $480/24 = 20$ units

The part of tank filled by pipe B alone in 1 minute = $480/16 = 30$ units

The part of tank emptied by pipe C alone in 1 minute = $480/32 = 15$ units

The part of tank emptied by pipe D alone in 1 minute = $480/40 = 12$ units

The part of tank filled by pipes B and C together in 4.4 minutes = $(30 - 15) * 4.4 = 15 * 4.4 = 66$ units

So, the part of tank filled by all the four pipes together = $480 - 66 = 414$ units

The required time taken = $414/(20 + 30 - 15 - 12) = 414/23 = 18$ minutes

So, pipes A and D should be closed after 18 minutes

16. Questions

Series I:

The given series follows the following pattern:

$$(11 * 1) + 1 = 12$$

$$(12 * 2) - 2 = 22$$

$$(22 * 3) + 3 = 69$$

$$(69 * 4) - 4 = 272$$

$$(272 * 5) + 5 = 1365$$

Since, both the series follow the same pattern.

@QuizByMickey

So,

$$(6 * 1) + 1 = 7$$

$$(7 * 2) - 2 = 12$$

$$(12 * 3) + 3 = 39$$

Since, $X = 7$

And, $X + Y = 12$

So, $Y = 12 - 7$

$$Y = 5$$

Since, $X + Y + Z = 39$

So, $Z = 39 - (7 + 5)$

$$Z = 27$$

Answer: A

From P:

The value of $(Z - 3X)^Y = (27 - (3 * 7))^5 = 6^5 = 7776$

So, P is not true.

From Q:

The value of $(Z - X - Y) = (27 - 7 - 5) = 15$

Thus, 15 is not a prime number.

So, Q is not true.

From R:

$$2(X + Y) < Z = 2(7 + 5) < 27$$

$$24 < 27$$

So, R is true.

Therefore, only R is true.

Hence, option A

17. Questions

Series I:

The given series follows the following pattern:

$$(11 * 1) + 1 = 12$$

$$(12 * 2) - 2 = 22$$

$$(22 * 3) + 3 = 69$$

@QuizByMickey

$$(69 * 4) - 4 = 272$$

$$(272 * 5) + 5 = 1365$$

Since, both the series follow the same pattern.

So,

$$(6 * 1) + 1 = 7$$

$$(7 * 2) - 2 = 12$$

$$(12 * 3) + 3 = 39$$

$$\text{Since, } X = 7$$

$$\text{And, } X + Y = 12$$

$$\text{So, } Y = 12 - 7$$

$$Y = 5$$

$$\text{Since, } X + Y + Z = 39$$

$$\text{So, } Z = 39 - (7 + 5)$$

$$Z = 27$$

Answer: C

$$\text{The 1}^{\text{st}} \text{ term of the series} = (Z - 2X - 2Y) = (27 - 14 - 10) = 3$$

$$\text{So, the 2}^{\text{nd}} \text{ term of the series} = (3 * 1) + 1 = 4$$

$$\text{The 3}^{\text{rd}} \text{ term of the series} = (4 * 2) - 2 = 6$$

$$\text{The 4}^{\text{th}} \text{ term of the series} = (6 * 3) + 3 = 21$$

$$\text{And the 5}^{\text{th}} \text{ term of the series} = (21 * 4) - 4 = 80$$

18. Questions

Series I:

The given series follows the following pattern:

$$(5 * 2) + 3 = 13$$

$$(13 * 2) + 3 = 29$$

$$(29 * 2) + 3 = 61$$

$$(61 * 2) + 3 = \mathbf{125}$$

$$(125 * 2) + 3 = 253$$

$$\text{So, the missing term of series I} = P = 125$$

Series II:

@QuizByMickey

The given series follows the following pattern:

$$1200 - 19^2 = 839$$

$$839 - 17^2 = \mathbf{550}$$

$$550 - 13^2 = 381$$

$$381 - 11^2 = 260$$

$$260 - 7^2 = 211$$

So, the missing term of series II = Q = 550

Series III:

The given series follows the following pattern:

$$10 * 3 = 30$$

$$30 * 2.5 = 75$$

$$75 * 2 = 150$$

$$150 * 1.5 = 225$$

$$225 * 1 = \mathbf{225}$$

So, the missing term of series III = R = 225

Answer: D

The ratio between P, Q and R = 125: 550: 225 = 5: 22: 9

19. Questions

Series I:

The given series follows the following pattern:

$$(5 * 2) + 3 = 13$$

$$(13 * 2) + 3 = 29$$

$$(29 * 2) + 3 = 61$$

$$(61 * 2) + 3 = \mathbf{125}$$

$$(125 * 2) + 3 = 253$$

So, the missing term of series I = P = 125

Series II:

The given series follows the following pattern:

$$1200 - 19^2 = 839$$

@QuizByMickey

$$839 - 17^2 = \mathbf{550}$$

$$550 - 13^2 = 381$$

$$381 - 11^2 = 260$$

$$260 - 7^2 = 211$$

So, the missing term of series II = Q = 550

Series III:

The given series follows the following pattern:

$$10 * 3 = 30$$

$$30 * 2.5 = 75$$

$$75 * 2 = 150$$

$$150 * 1.5 = 225$$

$$225 * 1 = \mathbf{225}$$

So, the missing term of series III = R = 225

Answer: E

$$\text{The value of } (Q - P^{1/3} - R^{1/2}) = (550 - (125)^{1/3} - (225)^{1/2}) = (550 - 5 - 15) = 530$$

Thus, 530 is divisible by neither 20, nor by 103, nor by 15.

So, option E is correct.

20. Questions

Answer: C

Series I:

The given series follows the following pattern:

21	36	57	90	141	216
+15		+21	+33	+51	+75
	+6	+12	+18	+24	

@QuizByMickey

So, the missing term in series I = P = 36

Series II:

The given series follows the following pattern:

$$21 + 23 = 44$$

$$44 + 29 = 73$$

$$73 + 31 = 104$$

$$104 + 37 = \mathbf{141}$$

$$141 + 41 = 182$$

So, the wrong term of series II = Q = 140

And the term, which will replace the wrong term of series II = R = 141

$$\text{The required average} = (36 + 140 + 141)/3 = 317/3 = 105.67$$

@QuizByMickey